Large grid subsets with no d+2 points on a sphere or hyperplane

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Allowed standard prerequisites (editorial).

- Standard VC-dimension, incidence bounds, Chernoff and Markov inequalities. Original author statements and references retained. The source-specific VC bound, cohyperplane count, sphere crossing observation and main random construction are included.

1 Introduction

The famous *no-three-in-line problem*, raised by Dudeney in 1917 in the special case $n = 8$, asks if it is possible to select $2n$ points from the lattice square $[n]^2$ such that no three are collinear. Clearly, one cannot do better than $2n$ as $[n]^2$ can be covered by n lines. For $n \leq 52$, many authors have published solutions to this problem obtaining the bound of $2n$ (e.g. see [6]). However, for large values of n, the best known lower bound is due to Hall et al. [8] which contains at least $\left(\frac{3}{2} - \varepsilon\right) n$ points, for any $\varepsilon > 0$ and n sufficiently large. The similar *no-four-on-a-circle* problem, raised by Erdős and Purdy in 1981 (see [13]), asks to determine the maximum number of points that can be selected from $[n]^2$ such that no four are on a circle. Here, a line is also considered to be a *degenerate* circle. Again, we have the trivial upper bound of $3n$ as any vertical line must contain at most three points. This upper bound was improved by Thiele [19], who showed that at most $\frac{5}{2}n - \frac{3}{2}$ points can be selected, and moreover, he gave a construction of $\frac{n}{4}$ points from $[n]^2$ with no four on a circle (or a line). In this paper, we study the no-four-on-a-circle problem in higher dimensions (Problem 4 in Chapter 10 of [4]). A *k-sphere* is a k-dimensional sphere. Thus, a 0-sphere is a pair of points, a 1-sphere is a circle, and etc. For simplicity, we will simply use the term *sphere* when referring to a $(d-1)$-dimensional sphere in $\mathbb{R}^d$. Again, the maximum number of points that can be selected from the d-dimensional lattice cube $[n]^d$ with no $d+2$ points on a sphere or a hyperplane is at most $(d+1)n$ since we can cover $[n]^d$ with n hyperplanes. In the other direction, Thiele [20] showed that one can select $\Omega(n^{\frac{1}{d-1}})$ points from $[n]^d$ with no $d+2$ points on a sphere or a hyperplane, providing the first non-trivial construction for this problem. In this note, we prove the following.

Theorem 1.1. *Let $d \geq 3$ be a positive integer. Then there is a subset of the d-dimensional lattice cube $[n]^d$ with $n^{\frac{3}{d+1}-o(1)}$ points with no $d+2$ members on a sphere or a hyperplane.*

While it is possible that one can improve this bound to $\Omega(n)$, we make the following more modest conjecture.

Conjecture 1.2. *Let $d \geq 3$ be a positive integer. Then there is a subset of the d-dimensional lattice cube $[n]^d$ with $\Omega(n^{\frac{d}{d+1}})$ points with no $d+2$ members on a sphere or a hyperplane.*

Our paper is organized as follows. In Section 2, we recall several results from VC-dimension theory that will be used in the proof of Theorem 1.1. In Section 3, we prove Theorem 1.1. We conclude our paper with several remarks and related problems. For sake of clarity, we systemically remove floor and ceilings whenever they are not crucial. All logarithms are in based 2 unless stated otherwise.

2 VC-dimension theory

In this section, we recall several results from VC-dimension theory that will be used in the proof of Theorem 1.1. First we introduce some terminology. Let $G = (P, Q, E)$ be a bipartite graph with independent sets P, Q and edges $E \subset P \times Q$. For $q \in Q$ we define the neighbourhood $N_Q(q) = \{p \in P \colon (p, q) \in E\}$. Define the set-system $\mathcal{F} = \{N_G(q) \colon q \in Q\}$ with ground set P. We say a set $S \subset P$ is *shattered* by $\mathcal{F}$ if for every subset $B \subset S$, there is a set $A \in \mathcal{F}$ such that $A \cap S = B$. The *Vapnik-Chervonenkis dimension* (VC-dimension) of $(P, \mathcal{F})$ is the largest integer d for which there exists a subset $S \subset P$, with $|S| = d$, that is shattered by $\mathcal{F}$. The *primal shatter function* of $(P, \mathcal{F})$ is defined as

$$\pi_{\mathcal{F}}(z) = \max_{P' \subset P, |P'| = z} |\{A \cap P' \colon A \in \mathcal{F}\}|.$$

The well-known Sauer-Shelah lemma [15, 17] states that if d_0 is the VC-dimension of $\mathcal{F}$, then

$$\pi_{\mathcal{F}}(z) \leq \sum_{i=0}^{d_0} \binom{z}{i}. \tag{1}$$

We will need the following result due to Fox, Pach, Sheffer, Suk, and Zahl.

Theorem 2.1 ([7])**.** *Let $G = (P, Q, E)$ be a bipartite graph with $|P| = m$ and $|Q| = n$, such that the set system $\mathcal{F}_1 = \{N(q) \colon q \in Q\}$, with ground set P, satisfies $\pi_{\mathcal{F}_1}(z) \leq cz^{d_0}$ for all z. Then if G is $K_{t,t}$-free, we have*

$$|E(G)| \leq c_1(mn^{1-1/d_0} + n),$$

where $c_1 = c_1(c, d_0, t)$. In particular, $c_1 \leq 10 \cdot ct^{2d_0+1}(d_0 \log d_0)^{d_0}$.

A subset $A \subset [n]^d$ is a *maximal spherical set of* $[n]^d$ if all points in A lie on a single sphere S in $\mathbb{R}^d$, and no point of $[n]^d$ can be added to A while keeping all points on the sphere S. Let $\mathcal{S}_{n,d}$ denote the set-system of all maximal spherical sets of $[n]^d$, with ground set $[n]^d$.

Lemma 2.2. *For $d \geq 2$, the VC-dimension of the set-system $([n]^d, \mathcal{S}_{n,d})$ is at most $d+1$.*

Proof. We proceed by induction on d. The base case $d = 1$ is trivial since a 0-sphere consists of 2 points. For the inductive step, assume the statement holds for $d' < d$. For sake of contradiction, suppose that a set Q of $d+2$ points in $[n]^d$ is shattered by $\mathcal{S}_{n,d}$. *Case 1.* Suppose Q is in general position, that is, no $d+1$ members in Q lie on a hyperplane. By linear algebra, any $d+1$ points in Q determines a unique sphere in $\mathbb{R}^d$. However, since Q is shattered by $\mathcal{S}_{n,d}$, there is a sphere that contains Q. Hence, there is no sphere that contains exactly $d+1$ points from Q. Contradiction. *Case 2.* Suppose Q contains a $(d+1)$-tuple Q' that lies on a hyperplane h in $\mathbb{R}^d$. Since the intersection of a sphere with h is a $(d-2)$-sphere in h, by induction on d, Q' cannot be shattered by $\mathcal{S}_{n,d}$. Hence, Q cannot be shattered by $\mathcal{S}_{n,d}$, which is a contradiction. □

3 Proof of Theorem 1.1

In this section, we prove Theorem 3. First, let us recall several results. The first is the well-known Chernoff inequality (see [10, Theorem 2.8]).

Lemma 3.1 (Chernoff's inequality)**.** *Let $X_1, \dots, X_n$ be independent random variables such that $\mathbb{P}(X_i = 1) = q$ and $\mathbb{P}(X_i = 0) = 1 - q$, and let $X = \sum_i X_i$. Then for $0 < \delta < 1$, we have*

$$\mathbb{P}(X \geq (1+\delta)qn) \leq e^{-\frac{\delta^2 qn}{3}},$$

and

$$\mathbb{P}(X \leq (1-\delta)qn) \leq e^{-\frac{\delta^2 qn}{3}}.$$

Next, we estimate the number of points from $[n]^d$ that can lie on a sphere or a hyperplane. For the latter, we will use the following result due to Balogh and White.

Lemma 3.2 ([2])**.** *Let $d \geq 2$ be a positive integer, and $a_1, \dots, a_d \in \mathbb{Z}$ not be all zero with greatest common denominator 1. Let $s = \max_i\{|a_i|\}$, $n \geq s$ be a positive integer, and set*

$$\mathcal{L} = \{(x_1, \dots, x_d) \in \mathbb{Z}^d \colon \sum_{i=1}^{d} a_i x_i = 0\}.$$

If $\mathcal{L} \cap [n]^d$ spans a $(d-1)$-dimensional subspace, then $|\mathcal{L} \cap [n]^d| \leq 3^d n^{d-1}/s$.

Lemma 3.3. *The number of $(d+2)$-tuples in $[n]^d$ that lie on a common hyperplane is at most $O(n^{d^2+d-1})$.*

Proof. By the lemma above, the number of $(d+1)$-tuples in $[n]^d$ that lie on a hyperplane which passes through the origin is at most

$$\sum_s \left(\frac{n^{d-1}}{s}\right)^{d+1} s^{d-1} = O(n^{d^2-1}).$$

By symmetry, for any fixed point $p \in [n]^d$, the number of $(d+1)$-tuples in $[n]^d$ that lies on a hyperplane which passes through p is at most $O(n^{d^2-1})$. After summing over all points in $[n]^d$, the statement follows. Note that Lemma 3.3 applies to hyperplanes whose intersection with $[n]^d$ spans a $(d-1)$-dimensional subspace. We can assume that any $(d+2)$-tuple in $[n]^d$ lying on a common hyperplane meets this requirement by adding other points from $[n]^d$ if needed. □

Finally, we will use the following result due to Sheffer that bounds the number of points from $[n]^d$ that lies on a k-sphere in $\mathbb{R}^d$, for $k \leq d-1$. See also [9, Sec. 11.2, Equation 11.9], [18, Theorem 2].

Lemma 3.4 ([16], Lemma 3.2)**.** *Let $d \geq 2$ and $k \leq d-1$. Then there is a positive constant $c = c(d)$ such that every k-sphere $\mathbb{R}^d$ contains at most $n^{k-1+c/\log\log n}$ points from $[n]^d$.*

Before we prove Theorem 1.1, let us give a brief outline of the argument. First, we use the probabilistic method and Lemmas 3.3 and 3.4 to obtain a large subset $A \subset [n]^d$ such that very few members of A lie on a $(d-2)$-sphere or a hyperplane in $\mathbb{R}^d$. We then apply Theorem 2.1 and ideas from incidence geometry to estimate the number of r-tuples in A that lie on a common sphere. Finally, we apply the deletion method to find a large subset $A'' \subset A$ such that no $d+2$ members of A'' lie on a sphere or a hyperplane. We now flesh out all of the details of the proof.

Proof of Theorem 1.1. Without loss of generality, we can assume that n is a power of 2. Indeed, otherwise we can find an integer $n' < n$ that is a power of 2 such that $n' > n/2$, and apply the arguments below to the subcube $[n']^d$. This would only change the hidden constant in the $o(1)$ term in the exponent. Moreover, we will assume that n is sufficiently large. Fix a positive integer $D < n$ that will be determined later, such that D divides n. We partition the d-dimensional cube $[n]^d$ into D^d smaller cubes $Q_1, \ldots, Q_{D^d}$, where each Q_i is of the form

$$\left\{i_1\frac{n}{D}+1, \ldots, (i_1+1)\frac{n}{D}\right\} \times \cdots \times \left\{i_d\frac{n}{D}+1, \ldots, (i_d+1)\frac{n}{D}\right\}.$$

where $i_1, \ldots, i_d \in \{0, \ldots, D-1\}$. Hence, $|Q_i| = (n/D)^d$. Consider a random subset $A \subset [n]^d$ obtained by selecting each point in $[n]^d$ independently with probability n^{3-d}. Set $P_i = Q_i \cap A$, for $i = 1, \ldots, D^d$. Let $\mathcal{W}$ be the event that the subset $A \subset [n]^d$ satisfies the following properties.

1. $n^3/2 \leq |A| \leq 2n^3$.
2. $n^3 D^{-d}/2 \leq |P_j| \leq 2n^3 D^{-d}$ for all $1 \leq j \leq D^d$.
3. Every $(d-2)$-sphere contains less than $n^{c_1/\log\log n}$ points of A, where $c_1 = c_1(d)$.
4. The number $(d+2)$-tuples in A that lie on a common hyperplane is at most $c_2 n^{2d+5}$, where $c_2 = c_2(d)$.

By Lemma 3.1 and Markov's inequality, event $\mathcal{W}$ holds with probability at least $1/2$. Indeed, by Lemma 3.1, the probability that $|A| > 2n^3$ or $|A| < n^3/2$ is at most $e^{-\frac{n^3}{3}} + e^{-\frac{n^3}{12}}$. Thus, the first property holds with high probability, that is, with probability tending to 1 as n approaches infinity. A similar argument follows for the second property. For the third property, let us fix a $(d-2)$-sphere S in $\mathbb{R}^d$. By Lemma 3.4, there is a constant $c = c(d)$ such that S contains at most $n^{d-3+c\log\log n}$ points from $[n]^d$. By Lemma 3.1,

$$\mathbb{P}[|S \cap A| \geq 2n^{c/\log\log n}] \leq e^{-\frac{n^{c/\log\log n}}{3}}.$$

Since there are at most n^{d^2} $(d-2)$-spheres in $\mathbb{R}^d$ with at least d points from $[n]^d$, the union bound implies that the third property holds with high probability. For the fourth property, let X denote the number $(d+2)$-tuples in A that lie on a common hyperplane. By Lemma 3.3, we have $\mathbb{E}[X] \leq c' n^{2d+5}$, where $c' = c'(d)$. By Markov's inequality (see [1]), we have

$$\mathbb{P}[X > 10c' n^{2d+5}] < \frac{\mathbb{E}[X]}{10c' n^{2d+5}} < \frac{1}{10}.$$

Putting everything together, and setting c_1, c_2 sufficiently large, event $\mathcal{W}$ holds with probability at least $1/2$. Thus, let us fix $A \subset [n]^d$ with the four properties described above, and set $t = 2n^{c_3/\log\log n}$, where $c_3 = \max\{c_1, c_2\}$. Let $\mathcal{S}$ be the collection of spheres in $\mathbb{R}^d$, such that each sphere in $\mathcal{S}$ contains at least $d+1$ points from A in general position. Hence, $|\mathcal{S}| = O(|A|^{d+1})$. Let $\mathcal{S}_j \subset \mathcal{S}$ be the set of spheres in $\mathcal{S}$ that contains at least one point of $P_j = Q_j \cap A$. Let $G_j = (P_j, \mathcal{S}_j, E_j)$ be the bipartite incidence graph between P_j and $\mathcal{S}_j$. Since the intersection of two distinct spheres in $\mathbb{R}^d$ is a $(d-2)$-sphere, and every $(d-2)$-sphere contains less than t points from A, each graph G_j is $K_{t,2}$-free. Let $\mathcal{F}_j$ be the set system whose ground set is P_j, and whose sets are $S \cap P_j$, where $S \in \mathcal{S}_j$ and $|S \cap P_j| \geq t$. That is,

$$\mathcal{F}_j = \{S \cap P_j : S \in \mathcal{S}_j, |S \cap P_j| \geq t\}.$$

By Lemma 2.2, the VC-dimension of $\mathcal{F}_j$ is at most $d+1$. By inequality (1), we have

$$\pi_{\mathcal{F}_j}(z) = O(z^{d+1}).$$

Hence, we apply Lemma 2.1 with $t = n^{c_3/\log\log n}$ to conclude that

$$|E_j| \leq n^{c_4/\log\log n}\left(|P_j||\mathcal{S}_j|^{\frac{d}{d+1}} + |\mathcal{S}_j|\right),$$

where $c_4 = c_4(d)$. Given the partition $[n]^d = Q_1 \cup \cdots \cup Q_{D^d}$ described above, we say that a sphere S *crosses* the subcube Q_i if $S \cap Q_i \neq \emptyset$.

Observation 3.5. *For fixed $d \geq 2$, every sphere S in $\mathbb{R}^d$ crosses at most $c_5 D^{d-1}$ subcubes Q_i, where $c_5 = c_5(d)$.*

Proof. Let $c_5 = c_5(d)$ be a large constant that depends only on d. We will determined c_5 later. For sake of contradiction, suppose there is a sphere S in $\mathbb{R}^d$ the crosses more than $c_5 D^{d-1}$ subcubes Q_i. Note that we will not make any serious attempts to optimize the constant c_5. Let $T \subset [n]^d \cap S$ be the subset of points $[n]^d \cap S$ obtained by selecting 1 point from each subcube Q_i that S crosses. Hence, $|T| \geq c_5 D^{d-1}$. Let $\mathcal{G} = (T, E)$ be an auxiliary graph on T, where two points in T are adjacent if their distance is less than n/D. Since two points in T have distance less than n/D only if they lie in adjacent subcubes Q_i and Q_j, this implies that $\mathcal{G}$ has maximum degree $3^d - 1$. By Turan's theorem, we can find a subset $T' \subset T$, such that the minimum distance between any two points in T' is at least n/D. Moreover, $|T'| \geq (c_5/3^d)D^{d-1}$. For each point $p \in T'$, consider the spherical cap C_p obtained by intersecting S with the ball B_p centered at p with radius $n/(2D)$. Since the minimum distance among the points in T' is at least n/D, the collection of spherical caps corresponding to the points in T' will have pairwise disjoint interiors. Moreover, since each spherical cap arises from a ball whose center is on S with radius $n/(2D)$, each spherical cap will have area at least $c_6(n/D)^{d-1}$ where $c_6 = c_6(d)$. Hence, the total area of all of the spherical caps corresponding to the points in T' is at least

$$|T'|c_5(n/D)^{d-1} \geq \frac{c_5 c_6}{3^d} n^{d-1}.$$

On the other hand, all of the spherical caps on S will lie inside of a "large" ball B with radius dn. Hence, the area of all of the spherical caps is at most the surface area of the ball B with radius dn. Since the surface area of B is at most $c_7 n^{d-1}$, where $c_7 = c_7(d)$, we have

$$c_7 n^{d-1} \geq \frac{c_5 c_6}{3^d} n^{d-1}.$$

By setting c_5 sufficiently large, we have a contradiction. □

By the observation above, $\sum_{j=1}^{D^d} |\mathcal{S}_j| = O(D^{d-1}|\mathcal{S}|)$. Putting all of these bounds together and

applying Jensen's inequality, the number of incidences between A and $\mathcal{S}$ is at most

$$\begin{aligned}
\sum_{j=1}^{D^d} |E_j| &\le n^{c_4/\log\log n} \sum_{j=1}^{D^d} \left(|P_j||\mathcal{S}_j|^{\frac{d}{d+1}} + |\mathcal{S}_j| \right) \\
&\le 2n^{3+c_4/\log\log n} D^{-d} \sum_{j=1}^{D^d} |\mathcal{S}_j|^{\frac{d}{d+1}} + n^{c_4/\log\log n} \sum_{j=1}^{D^d} |\mathcal{S}_j| \\
&\le 2n^{3+c_4/\log\log n} D^{-d} D^{\frac{d}{d+1}} \left(\sum_{j=1}^{D^d} |\mathcal{S}_j| \right)^{\frac{d}{d+1}} + c_5 D^{d-1} n^{c_4/\log\log n} |\mathcal{S}| \\
&\le 2c_5^{\frac{d}{d+1}} n^{3+c_4/\log\log n} D^{-\frac{d}{d+1}} |\mathcal{S}|^{\frac{d}{d+1}} + c_5 D^{d-1} n^{c_4/\log\log n} |\mathcal{S}|.
\end{aligned}$$

Set D to be a power of 2 such that

$$n^{\frac{3(d+1)}{d^2+d-1}} |\mathcal{S}|^{-\frac{1}{d^2+d-1}} \le D \le 2n^{\frac{3(d+1)}{d^2+d-1}} |\mathcal{S}|^{-\frac{1}{d^2+d-1}}.$$

Note that $|\mathcal{S}| \le |A|^{d+1}$, and therefore $1 < D < n$. From above, the number of incidences between $\mathcal{S}$ and A is at most

$$n^{\frac{3(d^2-1)}{d^2+d-1} + c_8/\log\log n} |\mathcal{S}|^{\frac{d^2}{d^2+d-1}},$$

where $c_8 = c_8(d)$. Let $\mathcal{S}_r \subset \mathcal{S}$ denote the set of r-rich spheres in $\mathcal{S}$, that is, the set of spheres in $\mathcal{S}$ with at least r points from A. Then we have

$$r|\mathcal{S}_r| \le n^{\frac{3(d^2-1)}{d^2+d-1} + c_8/\log\log n} |\mathcal{S}_r|^{\frac{d^2}{d^2+d-1}},$$

which implies

$$|\mathcal{S}_r| \le n^{3(d+1)+c_8/\log\log n} r^{-\frac{d^2+d-1}{d-1}}.$$

Let $\mathcal{T}_i \subset \mathcal{S}$ be the set of spheres in $\mathcal{S}$ with at least 2^i points from A, and at most 2^{i+1} from A. Hence

$$|\mathcal{T}_i| \le |\mathcal{S}_{2^i}| = n^{3(d+1)+c_8/\log\log n} 2^{-i\frac{d^2+d-1}{d-1}}.$$

Therefore, the number of $(d+2)$-tuples of A that lie on a common sphere $S \in \mathcal{S}$ is at most

$$\sum_{i=1}^{\log|A|} |\mathcal{T}_i| \binom{2^{i+1}}{d+2} \le n^{3(d+1)+c_8/\log\log n} \sum_{i=1}^{\log|A|} 2^{-i\frac{1}{d-1}} \le n^{3(d+1)+c_8/\log\log n}.$$

We now consider a random subset $A' \subset A$ obtained by selecting each point in A independently with probability p, where p will be determined later. Let X be the number of $(d+2)$-tuples of A' that lie on a common sphere or a hyperplane. Then by linearity of expectation, we have

$$\mathbb{E}[X] \le n^{3(d+1)+c_8/\log\log n} p^{d+2} + O(n^{2d+5}) p^{d+2} \le n^{3(d+1)+c_9/\log\log n} p^{d+2},$$

and

$$\mathbb{E}[|A'|] = p|A| \ge n^3 p/2,$$

where $c_9 = c_9(d)$. Hence, there is a constant $c_{10} = c_{10}(d)$ such that for $p = n^{-3d/(d+1)-c_{10}/\log\log n}$, we have $\mathbb{E}[X] < \mathbb{E}[|A'|]/4$. By Lemma 3.1, there is a subset $A' \subset A$ of size at least $p|A|/2$, such that A' contains at most $p|A|/4$ $(d+2)$-tuples on a sphere or a hyperplane. By deleting one point from each $(d+2)$-tuple on a sphere or a hyperplane, we obtain a subset $A'' \subset A'$ of size at least

$$|A|p/4 \geq n^{\frac{3}{d+1}-2c_{10}/\log\log n} = n^{\frac{3}{d+1}-o(1)},$$

such that no $d+2$ members in A'' lie on a sphere or a hyperplane. □

4 Concluding remarks

Several authors [14, 11, 3] observed that if n is prime, then by selecting n points from $[n]^d$ on the "modular moment curve" $(x, x^2, \ldots, x^d) \mod n$, for $x = 1, \ldots, n$, no $d+1$ points will lie on a hyperplane, and moreover, no $2d+1$ points lie on a sphere. Unfortunately, this construction may contain $(d+2)$-tuples on a sphere. Nevertheless, one can make the following simple observation.

Theorem 4.1. *Let $d \geq 2$ be a fixed positive integer. Then there is a subset of the d-dimensional lattice cube $[n]^d$ of size $\Omega(n)$ with no $d+1$ points on a hyperplane, and no $2d$ points on a sphere.*

Proof. Let n be prime, and let $A \subset [n]^d$ be points from the d-dimensional cube on the moment curve $(x, x^2, \ldots, x^d) \mod n$, for $1 \leq x \leq \lfloor n/(4d) \rfloor$. Hence, $|A| = \lfloor n/(4d) \rfloor$. For sake of contradiction, suppose there is a sphere S in $\mathbb{R}^d$, such that S contains $2d$ points from A. Note that this means S contains $2d$ points of $[n]^d$ when viewed as a sphere in $\mathbb{F}_n^d$. Let the center of $S \subset \mathbb{F}_n^d$ be $(c_1, \ldots, c_d)$ and radius r, such that S contains $2d$ points from A. Then we have $2d$ solutions $s_1, s_2, \ldots, s_{2d}$ to the equation

$$(x - c_1)^2 + (x^2 - c_2)^2 + \cdots (x^d - c_d)^2 - r^2 = 0 \mod n.$$

On the other hand, by the division algorithm, we have

$$(x - c_1)^2 + (x^2 - c_2)^2 + \cdots (x^d - c_d)^2 - r^2 = (x - s_1)(x - s_2)\cdots(x - s_{2d}) \mod n,$$

which implies that $s_1 + \cdots + s_{2d} = 0 \mod n$ as the coefficient of x^{2d-1} is $0 \mod n$. However, $s_i \leq n/(4d)$, which implies $s_1 + \cdots + s_{2d} \neq 0 \mod n$, contradiction. Hence, no $2d$ points in A lie on a sphere and no $d+1$ points lie on a hyperplane. If n is not prime, we apply Bertrand's Postulate to obtain a prime $n' < n$ such that $n' > n/2$, and apply the argument above to $[n']^d$. □

Another natural question is to determine the maximum number of points that can be selected from $[n]^d$ with no $d+2$ points on a sphere, but allowing many points on a hyperplane. Since each point from $[n]^d$ lies on a sphere centered at the origin with radius $\sqrt{t}$, where $t = 1, 2, \ldots, dn^2$, we have an upper bound of $d(d+1)n^2$ for this problem. Using Lemma 3.4 and the probabilistic method, one can show the existence of $n^{2-4/(d+1)-o(1)}$ points from $[n]^2 \subset [n]^d$ with no $d+2$ on a sphere.

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